

L1 — Equivalence Completion (Paper 9): Rephasing Redundancy Exhaustion under Admissible Redescription

Amos Jay Maley

0. Scope and Non-Negotiables

This paper is eliminative. It introduces no new primitives, no new ontology, no new anchors, and no new tensor load. All results are invariant under admissible redescription.

We work with fixed fermion content, fixed gauge representations, and fixed Yukawa operator existence. No additional phases, compensating fields, or patching mechanisms are permitted.

1. Apparent Phase Freedom

In broken-phase bookkeeping, fermion mass matrices admit arbitrary complex phases. These phases are often treated as independent parameters, suggesting a continuous rephasing freedom.

In gauge-phase bookkeeping, field rephasings are treated as basis redundancies, and operator content appears phase-invariant. This mismatch motivates an apparent surplus of physical phases.

2. Admissibility Envelope

Allowed redescriptions include:

- unitary field redefinitions preserving gauge-covariant kinetic structure,
- global and relative rephasings within fixed fermion multiplets,
- equivalent parameterizations preserving operator content.

Any phase choice that cannot be transported between skins without importing new structure is inadmissible.

3. Transport of Phase Data

Phases appearing in mass matrices arise from complex coefficients of admissible Yukawa operators. Transport from the broken phase to the gauge phase maps these phases to operator coefficients modulo admissible field rephasings.

All transportable phase information is therefore already encoded at the operator level.

4. Exhaustion of Rephasing Freedom

Assume additional independent rephasing parameters beyond those fixed by operator content. Transport to the gauge phase.

Such parameters cannot be represented without: (i) introducing new fermion fields, (ii) enlarging the Higgs sector, or (iii) adding non-gauge-invariant operators. Each option violates the declared load or admissibility constraints.

Therefore, all admissible rephasing freedom is exhausted by field redefinitions and operator coefficients. No residual independent phases survive transport.

5. Core Result

Theorem 1 (Rephasing Redundancy Exhaustion under Admissible Transport). *Fix the fermion content, gauge representations, and admissible Yukawa operator set. Under admissible redescription and global constraints (no new load, no patches), all rephasing freedom is redundant.*

No independent physical phases exist beyond those already encoded in admissible operator structure.

Corollary 1 (Phase Non-Independence). *Any purported additional phase parameter either corresponds to a removable redefinition or violates admissible transport.*

Corollary 2 (Skin Artifact of Phase Freedom). *The appearance of surplus physical phases is a bookkeeping artifact of broken-phase representation. Cross-skin admissibility collapses this illusion.*

6. Closure Statement

Rephasing does not introduce independent physical structure. All admissible phase data is fixed by operator content and exhausted by field redefinitions. No residual degrees of freedom survive admissible redescription.